Unit 2

Lesson 3

Innovators in Action.

A Safer World.

Time Frame: [90 min]

Lesson Objectives:

By the end of this lesson students will be able to:

- Create 3D models of room layouts.
- Design evacuation routes inside the home that account for obstacles, accessibility features, and sensor-triggered alerts (e.g., fire alarms, gas leak warnings).

Challenge questions of the lesson.

- How can creating a 3D model of a room layout help you identify the most efficient use of space while ensuring clear and safe pathways for movement?
- When designing an evacuation route, how can you ensure it remains accessible and safe for everyone, including people using wheelchairs or other mobility aids?

SEL Relation

- **Self-Awareness:** Recognizing their growing capabilities in complex system design, integration, and problem-solving.
- **Self-Management:** Practicing meticulous planning, organization, and persistent debugging when working on multi-component projects.
- **Social Awareness:** Understanding how integrated, multi-sensory alert systems can effectively cater to diverse user needs and significantly improve overall safety and accessibility.
- **Relationship Skills:** Collaborating effectively on complex circuit assembly and code integration, and providing constructive feedback to peers.
- **Responsible Decision-Making:** Considering the reliability, accuracy, and ethical implications of automated multi-sensory alerts in real-world scenarios.

Engage

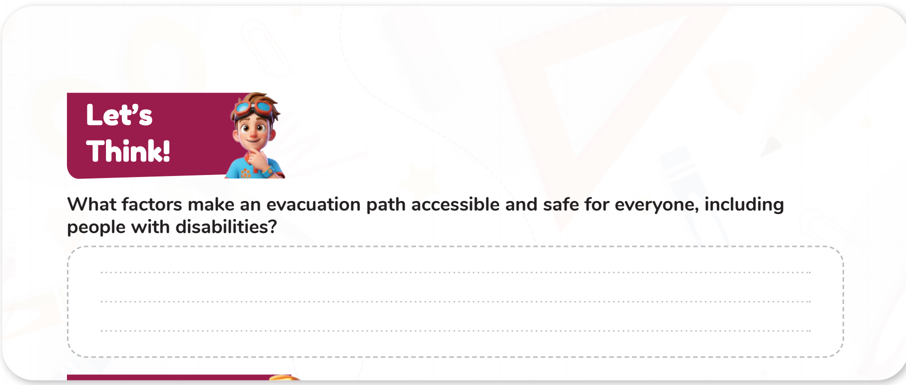
Teacher's Role:

- **Set the Scene:** Show a short video or story of an emergency, emphasizing challenges faced by people with impairments.
- **Ask Provocative Questions:** Use the question: "What factors make an evacuation path accessible and safe for everyone?"
- **Facilitate Empathy:** Ask students to imagine being in an emergency without hearing or sight. Optionally use blindfolds or headphones for empathy.
- **Transition to Today's Challenge:** Explain the task: students will design safe, accessible evacuation pathways.

Expected from Students:

- **Engage in Discussion:** Share ideas on safety and inclusivity in evacuation.
- **Ask Questions:** Explore how environment and disabilities affect evacuation planning.
- **Show Empathy:** Participate in activities and reflect on challenges faced by people with disabilities.
- **Get Inspired:** Show interest in solving real-world design problems for emergencies.

Parts of the book included.


Let's Think!

What factors make an evacuation path accessible and safe for everyone, including people with disabilities?

Let's Think!

Objective:

Students will explore key features for safe and accessible evacuation paths, including multi-sensory guidance, accessibility, and communication for people with disabilities during emergencies.

Answers:

- Clarity & Visibility: Well-lit paths, clear signage, no obstacles.
- Audibility: Loud alarms, clear tones, and instructions.
- Tactile Accessibility: Smooth surfaces, ramps, tactile paving, vibration alerts.
- Information & Guidance: Maps, assembly points, multi-format info.
- Accessibility Needs: Considerations for wheelchair users, visually/hearing impaired.
- Emergency Factors: Low-light conditions, smoke, crowd management.

Possible Strategies for Implementation:

1. **Mind Map Creation:**
   - Have students create mind maps on evacuation safety, linking ideas like vibration alerts and hearing impairments.
2. **Scenario-Based Brainstorming:**
   - Discuss how different individuals (e.g., wheelchair users, deaf people) would navigate evacuation paths.
3. **"5 Senses" Brainstorming:**
   - Explore how sensory inputs (sight, sound, touch, smell) can be used in evacuation paths.
4. **"Worst Case Scenario" Brainstorming:**
   - Identify unsafe evacuation factors and brainstorm solutions to make paths safer.

Explorer's Toolkit

Teachers' Role:

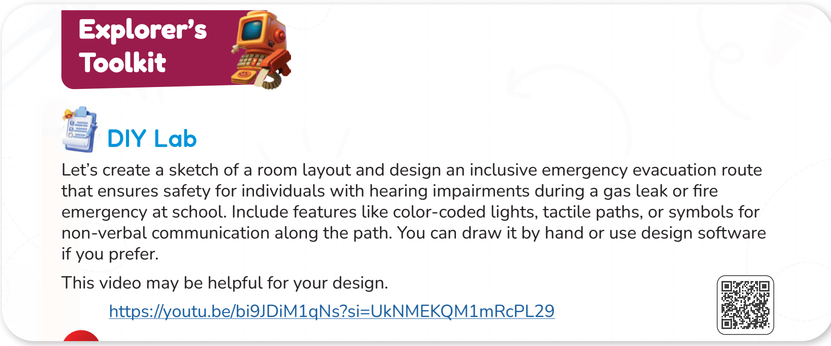
- Introduce the concept of inclusive evacuation route design, emphasizing the importance of accessibility for people with hearing impairments.
- Explain the safety features that could be included, such as color-coded lights, tactile floor paths, and non-verbal communication symbols.
- Show the provided video link and highlight key takeaways that could inspire student designs.

- Encourage creative problem-solving and remind students to consider both safety and accessibility.

Expected from Students:

- Create a sketch or 3D model of a room layout that includes an inclusive evacuation route.
- Integrate accessibility features (e.g., lights, tactile paths, symbols) into their design.
- Consider possible hazards and how to ensure the route remains clear during an emergency.
- Present the design, explaining how it meets both safety and accessibility needs.

Parts of the book included.



Explorer's Toolkit

DIY Lab

Let's create a sketch of a room layout and design an inclusive emergency evacuation route that ensures safety for individuals with hearing impairments during a gas leak or fire emergency at school. Include features like color-coded lights, tactile paths, or symbols for non-verbal communication along the path. You can draw it by hand or use design software if you prefer.

This video may be helpful for your design.

https://youtu.be/bi9JDiM1qNs?si=UkNMEKQM1mRcPL29

DIY Lab

Objective:

Students will design an inclusive emergency evacuation route that ensures safety and accessibility for individuals with hearing impairments during emergencies, applying principles of universal design and hazard planning.

Activity Link:

Video link: https://youtu.be/bi9JDiM1qNs?si=UkNMEKQM1mRcPL29

Software link: https://evacuation-planner.com/

Activity Instructions:

1. Watch the provided video to gather ideas about inclusive and accessible design.
2. Decide on the type of room you will design (e.g., classroom, office, lab).
3. Sketch your room layout, clearly marking pathways, exits, and sensor placements.
4. Add accessibility features such as:
   - Color-coded lights to guide visually from room to exit.
   - Tactile paths for navigation without sound.
   - Symbols or icons to convey instructions without verbal communication.
5. Ensure your evacuation route avoids common hazards and provides an alternate exit.
6. Submit your final design with a short explanation of your choices.

Explain

Teachers' Role:

- Introduce the purpose of this phase: to deepen understanding of emergency evacuation procedures, the role of evacuation maps, and inclusive planning for persons with disabilities.
- Connect both activities to prior safety knowledge and to real-world application in students' communities.
- Preview vocabulary that will appear in both the video and article (e.g., evacuation route, accessible exit, assembly point, visual alarms, mobility assistance).
- Guide active engagement with each resource by setting clear purposes, providing note-taking tools, and pausing for discussion when needed.
- Facilitate reflection and synthesis after both activities so students can identify general safety principles and specialized inclusive practices.

Expected from Students:

- Engage attentively with each resource, taking notes on key ideas.
- Identify the main purpose and important details of each activity.
- Compare and contrast general evacuation strategies with those tailored for people with disabilities.
- Participate in class discussions, contributing examples from the resources and personal observations.
- Reflect on how inclusive evacuation planning benefits everyone.

Parts of the book included.

Screen Lessons

Let's watch this video to understand how evacuation maps are important!

https://www.youtube.com/watch?v=3aLWlDY_G9w

Read it

Now! Let's read this article to learn more about Emergency evacuation for persons with disabilities.

https://mualert.missouri.edu/disabilities?utm_source=chatgpt.com

Screen Lessons

Objective:

- Understand the purpose of evacuation maps in emergency preparedness.
- Identify critical features that make evacuation maps effective and easy to follow.
- Explain how clear signage and route planning improve safety in emergencies.

Resources:

Link: https://www.youtube.com/watch?v=3aLWlDY_G9w

Possible Strategies for Implementation:

1. **Pre-Watching:**
   - Ask students: "Why do buildings have evacuation maps posted?"
   - Encourage sharing of experiences seeing these maps in places like schools, malls, or airports.
   - Explain that the video will demonstrate how evacuation maps can save lives during emergencies such as fires or gas leaks.
2. **During Watching:**
   - Instruct students to jot down 2–3 features that make an evacuation map effective.
   - Focus their attention with questions such as:
     - o "What hazards should be shown on a map?"
     - o "Who benefits from evacuation maps?"
3. **Post-Watching Discussion :**
   - Discuss key features every evacuation map should include (exits, hazard zones, escape routes, emergency equipment, clear symbols).
   - Explore improvements for accessibility for people with hearing or vision impairments (bold colors, large icons, tactile maps, lights, vibration signals).
   - Create a class list of "Must-Have Map Features" including exits, symbols, color coding, and hazard zones.

Dig Deeper

Objective:

Students will explore best practices and guidelines for emergency evacuation procedures designed specifically for individuals with disabilities, deepening their understanding of inclusive safety planning.

Activity Link:

https://mualert.missouri.edu/disabilities?utm_source=chatgpt.com

Activity Instructions:

1. Open the provided link and read the article carefully.

2. Take notes on at least three key points that are important for creating an inclusive evacuation plan.
3. Identify any specific challenges faced by individuals with disabilities during emergencies.
4. Consider how these insights can be applied to your own room layout and evacuation route design from the previous activity.
5. Be prepared to share your key takeaways in a class discussion or written reflection.

Possible Strategies for Implementation:

1. **Pre-Reading Discussion**
   - Begin with a short class discussion on why inclusive evacuation plans are important.
   - Ask students to brainstorm what challenges people with disabilities might face during an emergency.
2. **Guided Reading**
   - Have students read the article individually or in pairs, highlighting or noting important points.
   - Provide a worksheet with prompts such as "One thing I learned is...", "A challenge mentioned is...", and "A possible solution is...".
3. **Think-Pair-Share**
   - Students first jot down their own notes.
   - Then, pair up to share what they found before presenting to the larger group.
4. **Integration with Previous Activity**
   - Ask students to revisit their room layout and evacuation route design from the previous Explore task.
   - Instruct them to add or modify features based on insights gained from the article.
5. **Small-Group Roleplay**
   - Assign roles (e.g., person with a mobility aid, person with a hearing impairment, emergency guide).
   - Have students simulate an evacuation to see if their plan meets different needs.
6. **Exit Ticket Reflection**
   - At the end of the session, students write down one new idea they learned that they will apply to their design.

Elaborate

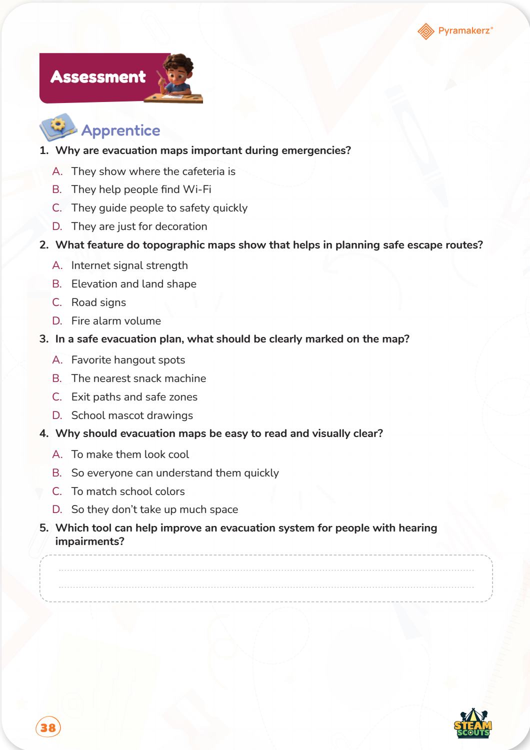
Teacher's Role:

- **Facilitate the transition:** Encourage students to go from understanding evacuation strategies to designing practical applications such as models (DIY Room) and tools (Evacuation Guidebook).
- **Provide resources and support:** Ensure students have the materials they need for their tasks (e.g., crafting tools for DIY Room model, templates for the guidebook).

Expected from students:

- **Apply Knowledge:**
  - Use their understanding of evacuation maps and accessibility to design practical solutions.
- **Complete the Apprentice Assessment:**
  - Answer questions about evacuation maps and accessible escape routes.
- **Build a DIY Room Model:**
  - Create a simple room model with doors, windows, and space for evacuation routes.
- **Collaborate and Share Ideas:**
  - Work with peers to discuss room layouts and share feedback on evacuation plans.
- **Design an Evacuation Guidebook:**
  - Create a guidebook with evacuation routes, emergency contacts, and accessibility tips.
- **Ensure Accessibility and Safety:**
  - Make sure their designs are safe and work for everyone, including people with disabilities.
- **Present and Reflect on Designs:**
  - Present their models and guidebooks, reflecting on what worked and what could be improved.

Parts of the book included.


Pyramakerz

Assessment

Apprentice

1. Why are evacuation maps important during emergencies?
   - A. They show where the cafeteria is
   - B. They help people find Wi-Fi
   - C. They guide people to safety quickly
   - D. They are just for decoration
2. What feature do topographic maps show that helps in planning safe escape routes?
   - A. Internet signal strength
   - B. Elevation and land shape
   - C. Road signs
   - D. Fire alarm volume
3. In a safe evacuation plan, what should be clearly marked on the map?
   - A. Favorite hangout spots
   - B. The nearest snack machine
   - C. Exit paths and safe zones
   - D. School mascot drawings
4. Why should evacuation maps be easy to read and visually clear?
   - A. To make them look cool
   - B. So everyone can understand them quickly
   - C. To match school colors
   - D. So they don't take up much space
5. Which tool can help improve an evacuation system for people with hearing impairments?

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STEAM SCOUTS


Builder

Now it's time to create your DIY room model, so you can later add your evacuation route to it.

Figure 14: DIY Room Model for Evacuation Planning.

Mastermind

Design & create a mini evacuation guidebook for everyone in your neighbourhood, which explains how to evacuate safely during emergencies.

It should include:

visual signs- symbols- tips for using apps for alerts- evacuation maps- emergency contact numbers.

Don't forget, it should be inclusive for everyone.

Figure 15: Evacuation guide book poster.

STEAM SCOUTS

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Apprentice

Objective:

Students will demonstrate their understanding of the importance of evacuation maps, the features of topographic maps for safe escape planning, key elements in evacuation plans, and tools that improve accessibility for people with hearing impairments. This assessment evaluates their grasp of map clarity, inclusivity, and emergency preparedness concepts.

Answers:

1. **Why are evacuation maps important during emergencies?**

C. They guide people to safety quickly

2. **What feature do topographic maps show that helps in planning safe escape routes?**

B. Elevation and land shape

3. **In a safe evacuation plan, what should be clearly marked on the map?**

C. Exit paths and safe zones

4. **Why should evacuation maps be easy to read and visually clear?**

B. So everyone can understand them quickly

5. **Which tool can help improve an evacuation system for people with hearing impairments?**

A helpful tool to improve an evacuation system for people with hearing impairments is a visual alarm system — for example, flashing light alarms or vibration alert devices connected to emergency systems.

These tools ensure that even if someone cannot hear sirens, they can still notice the alert through light or vibration signals and evacuate safely.

Possible Strategies for Implementation:

1. **Clarify Question Vocabulary:**

Guide students to identify key terms in each question (e.g., "evacuation maps," "topographic maps," "safe zones") before answering to ensure comprehension.

2. **Eliminate Incorrect Options:**

Encourage students to think critically and rule out answers that are clearly unrelated to emergency safety (e.g., decorations, school mascot drawings).

3. **Connect to Prior Learning:**

Remind students of previous lessons on evacuation map features, sensor technology, and inclusivity to reinforce concepts relevant to the questions.

4. **Provide Examples and Context:**

Use real-world examples or scenarios (e.g., fire drills, building evacuations) to contextualize questions and support student understanding.

5. **Encourage Reflection:**

After the quiz, facilitate a discussion where students explain their answers to deepen comprehension and address any misconceptions.

Builder

Objective:

- Apply spatial reasoning skills to design and create a physical model of a room.
- Understand the importance of room layout and accessibility in emergency evacuation planning.
- Identify key features (such as doors, windows, and furniture) that might influence evacuation routes.
- Prepare a foundation for adding accessible and effective evacuation routes in the next phase of the project.
- Enhance creativity by designing a room that can later accommodate evacuation routes, with a focus on safety and accessibility.

Materials:

- Cardboard or construction paper (for creating the walls and floor of the room)
- Scissors (for cutting the materials to size)
- Rulers (to measure and create straight lines)
- Markers or pens (to draw doors, windows, and labels on the room model)
- Glue or tape (to attach walls and any other components)
- Optional:

- o Small objects (such as figurines, blocks, or small furniture pieces) for representing furniture and other room features
- o Colored paper (to create furniture, windows, or other room details)
- o Stickers (for marking exits, doors, or emergency equipment

Activity Instructions:

1. **Step 1: Choose Your Room Type**

Decide on the type of room you want to model. It could be a classroom, a living room, an office, or any other room where people might need an evacuation plan.

- Tip: Consider the size of the room and the number of exits it might have. This will help you plan the evacuation route later.

2. **Step 2: Gather Materials**

Collect your materials. You will need:

- Cardboard or construction paper (for walls and floor)
- Markers, rulers, and scissors
- Tape or glue
- Any small objects you might want to use as furniture (optional)

3. **Step 3: Build Your Room Model**
   - Start by constructing the base (floor) and walls of your room. Use the ruler to make straight lines for accurate dimensions.
   - Add doors and windows where they would be in the real room. Ensure there is enough space for exits and possible evacuation routes.
   - If you want, you can add furniture or any other features that might impact the evacuation route.
4. **Step 4: Prepare for Evacuation Routes**
   - Once your room model is complete, think about where evacuation routes could go. Consider things like door placement, furniture arrangement, and any obstacles that might be in the way.
   - Leave space in the model to add your evacuation route in the next phase.
5. **Step 5: Final Check**

Ensure your model is sturdy, and all pieces are securely attached. This will make it easier to add the evacuation route in the next part of the project.

Possible Strategies for Implementation:

- **Design Assistance:** If students need help with their designs, encourage them to sketch their room layout on paper first. This can help with placement before they start building the model.
- **Visualization:** Before students begin cutting or gluing, have them visualize the room's layout and plan out where the evacuation route might go.
- **Collaboration:** Encourage students to work together when possible. They can share

ideas on how best to position doors and other features for a smooth evacuation route.

- **Accessibility Focus:** Remind students that evacuation routes should be accessible to everyone. How can the routes be optimized for people with mobility challenges or other disabilities?

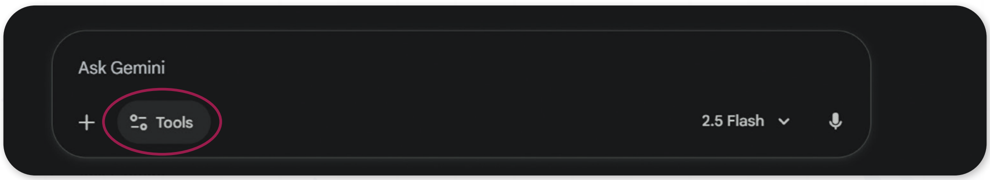
Mastermind

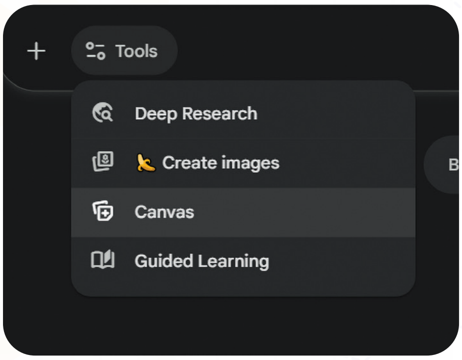
Objective:

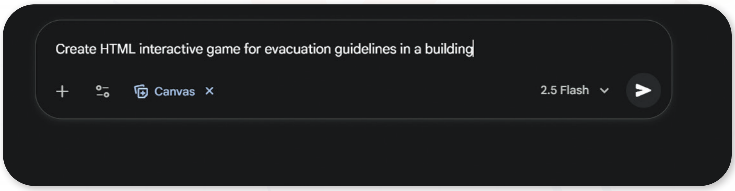
Students will design and create a mini evacuation guidebook for their neighborhood that clearly explains safe evacuation procedures during emergencies. The guidebook will incorporate visual signs, symbols, tips for using alert apps, evacuation maps, and emergency contact information to ensure accessibility and safety for all community members.

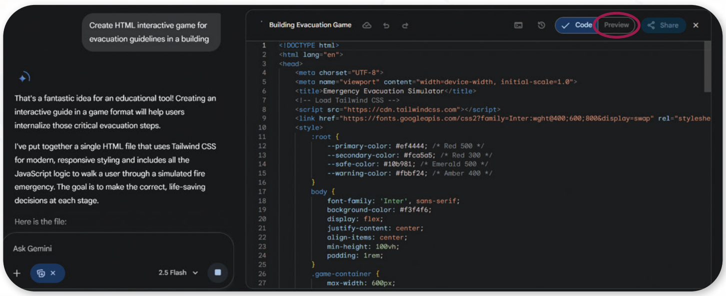
Activity Instructions:

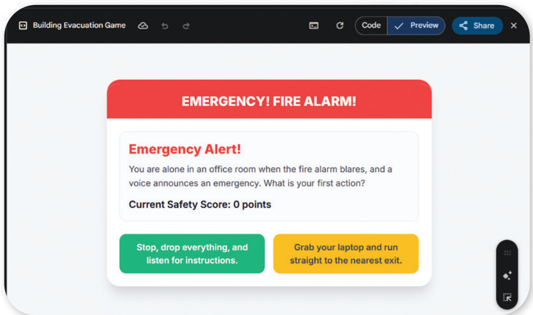
1. **Introduce the Task in Class**
   - The teacher introduces the project and explains its purpose: creating a mini evacuation guidebook for the local neighborhood.
   - Emphasize inclusivity by addressing the needs of children, the elderly, and people with disabilities.
   - Discuss examples of safety guidebooks used in workplaces and communities to build real-world understanding.
2. **Explain the Required Components (During Class)**
   - Visual signs and symbols for safety.
   - Tips for using mobile apps for emergency alerts.
   - Evacuation maps and emergency contacts.
   - Accessibility features such as color codes or tactile/visual indicators.
3. **AI Integration (Optional Extension)**
   - Demonstrate how to use Gemini Canva: https://gemini.google.com/app
   - Explain how students can use the tool to create an interactive HTML evacuation guide.

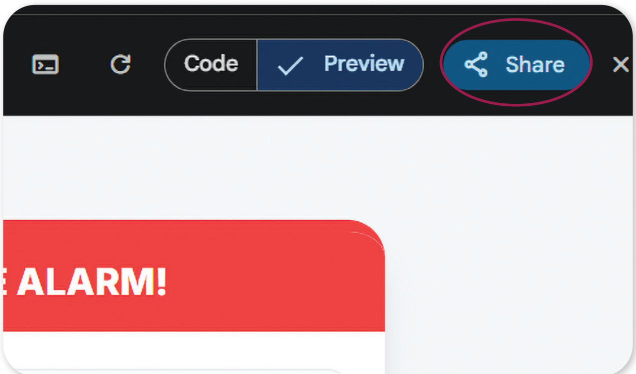


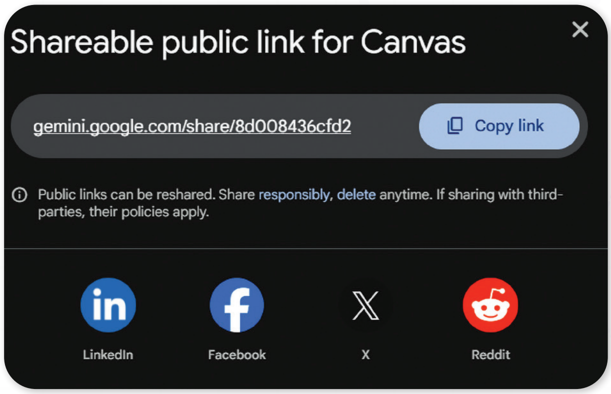












4. **Independent Work (At Home)**

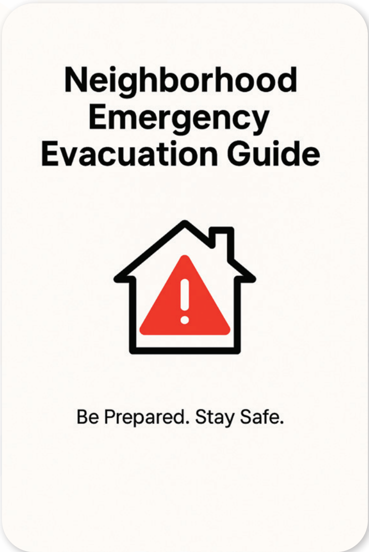
- Students complete their guidebooks individually at home.
- They may choose to create the guide digitally (Word, Canva, or Gemini Canva) or by hand, then digitize it later.
- Encourage them to research official safety websites for accurate emergency procedures.

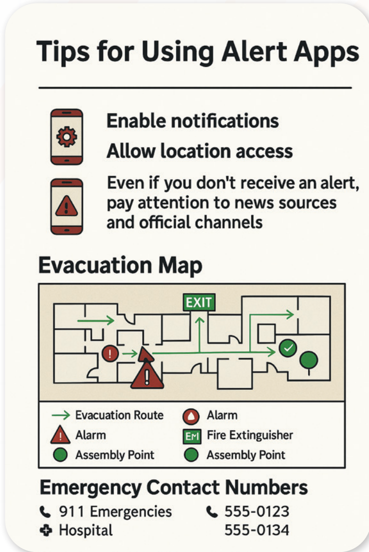
5. **Submission and Presentation (Next Class)**

- Students bring their completed guidebooks to class for a Showcase Activity.
- Each student presents their guide, explaining their design choices and how AI enhanced their project.

Sample Visual Concept for Mini Evacuation Guidebook

Here is a simple structure idea you can share with students:





Game link: https://gemini.google.com/share/745575f8e294

Evaluate

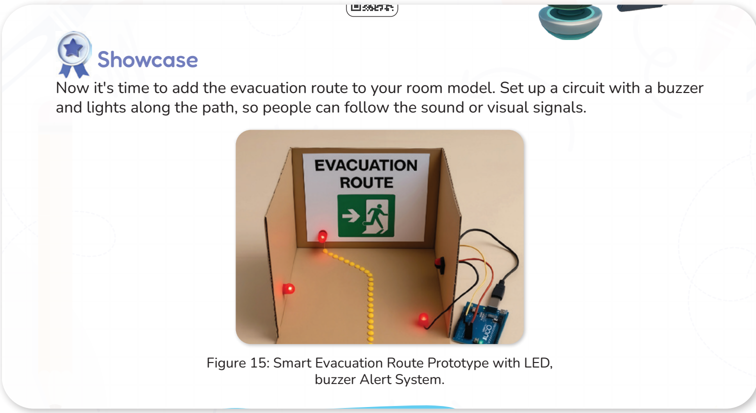
Teacher's Role:

- **Objective:** Guide students in evaluating the effectiveness of their evacuation route by using circuits to create auditory and visual alerts. Students will learn how to enhance their models with functional electronics that can guide people through an evacuation route.
- **Facilitate hands-on learning:** Ensure students understand how to set up and test the circuit, explaining how the buzzer and lights can help make evacuation routes more noticeable.
- **Encourage creativity:** Support students in adapting the circuit and designing the route in a way that makes it both practical and accessible, considering various needs (e.g., hearing-impaired individuals).
- **Assess student progress:** Observe and assess the students' ability to integrate their circuits into the model and explain the functionality of their evacuation route.

Expected from Students:

- **Integrate a buzzer and lights:** Students will set up a basic circuit with a buzzer and lights along their evacuation route in their room model.
- **Ensure functionality:** The circuit should be connected correctly, with lights and the buzzer functioning as intended to signal the evacuation route.
- **Demonstrate understanding:** Students should be able to explain how the buzzer and lights help guide people, especially those with disabilities, through the evacuation route.
- **Reflect on design choices:** Students will reflect on how their circuit can be used effectively in real-world emergencies and suggest any improvements or changes they might make.

Parts of the book Included.


Showcase

Now it's time to add the evacuation route to your room model. Set up a circuit with a buzzer and lights along the path, so people can follow the sound or visual signals.

Figure 15: Smart Evacuation Route Prototype with LED, buzzer Alert System.

Showcase

Objective:

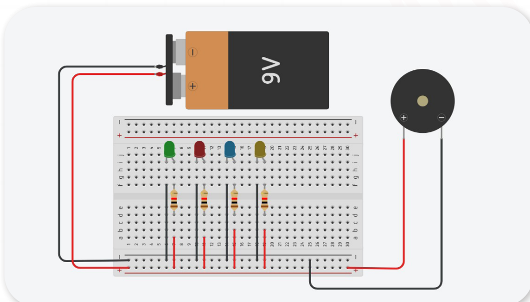
Design and integrate a buzzer and lights into your evacuation route to help people follow the path during an emergency. The circuit will use sound and visual signals to guide individuals to safety.

Required Materials:

- Breadboard
- Buzzer
- LED lights
- Wires and resistors
- Battery or power source
- Optional: Switches to control the circuit
- Room model (already built in previous tasks)

Activity Instructions:

1. **Set up the circuit:**
   - Place the LED lights along the path of your evacuation route on the room model.
   - Connect the buzzer at a starting or critical point on the route.



2. **Wire the circuit:**
   - Connect the lights and buzzer to a power source (e.g., a battery or power supply).
   - Use the breadboard to help make the connections, and ensure everything is wired safely.
3. **Test the circuit:**
   - Turn on the circuit and make sure the buzzer sounds, and the lights light up along the path when activated.

- If needed, adjust the circuit to ensure all elements are working properly.

4. **Reflect on functionality:**

- Think about how this alert system could be helpful during an emergency. Consider how it could be improved to work for people with disabilities.

Possible Strategies for Implementation:

- **Modeling the Circuit:** Demonstrate how to set up the basic circuit using LED lights and a buzzer on a simple breadboard before students start.
- **Pair Work:** Have students work in pairs to encourage collaboration and problem-solving while assembling their circuits.
- **Testing and Troubleshooting:** Encourage students to test their circuits frequently and troubleshoot any issues with the lights or buzzer.
- **Safety Considerations:** Remind students to be cautious when handling electrical components, ensuring all connections are secure to prevent short circuits.

Rubric with Clear Criteria

Criteria	Needs Improvement (1–2)	Satisfactory (3)	Excellent (4–5)
Functionality of Circuit	The buzzer or lights do not work, or the circuit is incomplete.	The buzzer and lights work, but the circuit needs some adjustments.	The circuit is fully functional, with both the buzzer and lights working as intended.
Creativity and Design	Limited creativity in the design of the evacuation route.	The evacuation route is functional, but lacks creative elements.	Highly creative design with a well-thought-out evacuation route and effective use of sound and light.
Clarity of Explanation	Unable to explain how the circuit works or its purpose.	Can explain the basic functionality of the circuit.	Clearly explains how the circuit works, its purpose in guiding evacuation, and how it can be improved.
Reflection and Improvement	Minimal reflection on the design process or potential improvements.	Some reflection on the design, with suggestions for improvement.	Thoughtful reflection on the design process with specific ideas for future improvements.

It’s time to move to the improve part and add this:

Improve

In the Improve step of the Engineering Design Process, we tested and enhanced our sound path by adding real circuits with buttons and buzzers. We looked back at our plans, checked where we wanted the sound buttons to go, and built working circuits to match our design. This made our playground safer and more helpful for children who can’t see well. By building, testing, and adjusting where we placed each button and buzzer, we made sure the path worked well and was easy to follow.

Now I Can

Objective:

Students will reflect on their learning and identify how they’ve grown in understanding communication, empathy, and the role of assistive tools in helping others share ideas.

Activity Instructions:

1. **Introduce the Reflection**
   - Say: “Let’s look back on everything we’ve explored this lesson. You learned how it feels to face communication challenges, and how tools can help people share their thoughts in special ways.”
2. **Guide Students Through the Checklist**
   - Read each statement aloud one at a time.
   - After each one, ask students to give a thumbs up, sideways, or down to show how confident they feel.
   - Example prompts:
     - “Did you learn how people share ideas without talking?”
     - “Do you know how tools can help people talk?”
3. **Student Completion**
   - Let students check off or color the “Now I Can” boxes they feel good about.
   - Remind them: “It’s okay if you didn’t check all of them. We’re all still learning.”

4. **Optional Extension**
   - Ask students to circle the one they're most proud of.
   - Invite them to write a sentence or draw a picture to show their favorite learning moment.

Relations to Standards:

1. Next Generation Science Standards (NGSS):
   - **3-5-ETS1-1:** Define a simple design problem reflecting a need or a want that includes specified criteria for success and constraints on materials, time, or cost.
   - **4-PS3-4:** Apply scientific ideas to design, test, and refine a device that converts energy from one form to another (e.g., electrical energy activating light, sound, or vibration).
   - **3-PS2-1:** Plan and conduct investigations to provide evidence of the effects of balanced and unbalanced forces on the motion of an object — supporting understanding of how sensors detect and respond to changes.
2. International Society for Technology in Education (ISTE) Standards for Students:
   - **Empowered Learner:** Use technology to take an active role in choosing, achieving, and demonstrating competency in their learning goals.
   - **Innovative Designer:** Use a variety of technologies within a design process to identify and solve problems by creating new, useful, or imaginative solutions.
   - **Creative Communicator:** Communicate clearly and express themselves creatively for a variety of purposes using digital tools.
3. Common Core State Standards (CCSS) — English Language Arts:
   - **CCSS.ELA-LITERACY.SL.3.1:** Engage effectively in collaborative discussions on grade-appropriate topics and texts, building on others' ideas and expressing their own clearly.
   - **CCSS.ELA-LITERACY.W.3.2:** Write informative/explanatory texts to examine a topic and convey ideas clearly.
   - **CCSS.MATH.PRACTICE.MP4:** Model with mathematics — interpret data and apply conditions logically in designing alert systems.

1. Social-Emotional Learning (SEL):
   - **Self-Awareness & Responsible Decision-Making:** Understanding the importance of creating inclusive and reliable alert systems that protect and assist all users, including those with sensory impairments.